SOMAIYA
VIDYAVIHAR
K J Somaiya Institute of Technology
An Autonomous Institute Permanently Affiliated to the University of Mumbai

Heer Haresh Panchal BS464

EXPERIMENT 1

Aim: Implementation of Stack using Array for real-world application.

Objective: To use stack operations such as Push, Pop, and Display to manage active heatwave alerts based on temperature readings.

Software Used: DOSBox, Turbo C ++

Theory

A stack is a linear data structure that follows the LIFO (Last In, First Out) principle, where the element inserted last is removed first. In this experiment, a stack is implemented using an array and a top variable is used to keep track of the topmost element.

The stack supports three basic operations: Push to insert an alert, Pop to remove the most recent alert, and Display to show all active alerts.

For the real-world application, the stack is used for heatwave alert prediction based on temperature:·

- **Temperature >= 45°C: Red Alert**
- **Temperature >=43°C: Orange Alert**
- **Temperature >= 40°C: Yellow Alert**
- **Temperature < 40°C: All active alerts are removed using Pop operations.**

The system provides a simple way to monitor temperature levels and manage active heatwave alerts. It demonstrates how stack operations can be applied to a real-world alert management system.

Program

```c
#include <stdio.h>
int stack[100], top = -1;

void Push(int alert)
{
    top++;
    stack[top] = alert;
}

void Pop() {
    if(top == -1)
    {
        printf("\nNo active alerts.");
    }
    else
    {
        if(stack[top] == 1)
            printf("\nPopped: Yellow Heatwave Alert");
        else if(stack[top] == 2)
            printf("\nPopped: Orange Heatwave Alert");
        else if(stack[top] == 3)
            printf("\nPopped: Red Heatwave Alert");

        top--;
    }
}

void Display()
{
    int i;

    if(top == -1)
    {
        printf("\nNo active alerts.");
    }
    else
    {
        printf("\nActive Heatwave Alerts:");

        for(i = top; i >= 0; i--)
        {
            if(stack[i] == 1)
                printf("\nYellow Heatwave Alert");
            else if(stack[i] == 2)
                printf("\nOrange Heatwave Alert");
            else if(stack[i] == 3)
                printf("\nRed Heatwave Alert");
        }
    }
}
```

```c
int main() {
    int choice;
    float temperature;
    do{
        printf("\n\nHEATWAVE ALERT STACK SYSTEM");
        printf("\n1. Enter Temperature");
        printf("\n2. Pop Alert");
        printf("\n3. Display Alerts");
        printf("\n4. Exit");
        printf("\nEnter your choice: ");
        scanf("%d", &choice);

        switch(choice){
            case 1:
                printf("\nEnter Temperature (in Celsius): ");
                scanf("%f", &temperature);
                if(temperature >= 40 && temperature < 43) {
                    printf("\nYellow Heatwave Alert");
                    Push(1);
                }
                if(temperature >= 43 && temperature < 45) {
                    printf("\nOrange Heatwave Alert");
                    Push(2);
                }
                if(temperature >= 45) {
                    printf("\nRed Heatwave Alert");
                    Push(3);
                }
                if(temperature < 40) {
                    printf("\nTemperature is Normal");
                    printf("\nRemoving all active alerts...");
                    while(top != -1){
                        Pop();
                    }}
                break;
            case 2:
                Pop();
                break;
            case 3:
                Display();
                break;
            case 4:
                printf("\nProgram Finished.");
                break;
            default:
                printf("\nInvalid Choice.");
        }
    } while(choice != 4);
    return 0;
}
```

Output

```
HEATWAVE ALERT STACK SYSTEM
1. Enter Temperature
2. Pop Alert
3. Display Alerts
4. Exit
Enter your choice: 1

Enter Temperature (in Celsius): 42

Yellow Heatwave Alert

HEATWAVE ALERT STACK SYSTEM
1. Enter Temperature
2. Pop Alert
3. Display Alerts
4. Exit
Enter your choice: 2

Popped: Yellow Heatwave Alert

HEATWAVE ALERT STACK SYSTEM
1. Enter Temperature
2. Pop Alert
3. Display Alerts
4. Exit
Enter your choice: 3

Active Heatwave Alerts:
Orange Heatwave Alert
Red Heatwave Alert

HEATWAVE ALERT STACK SYSTEM
1. Enter Temperature
2. Pop Alert
3. Display Alerts
4. Exit
Enter your choice: 1

Enter Temperature (in Celsius): 30

Temperature is Normal
Removing all active alerts...
Popped: Orange Heatwave Alert
Popped: Red Heatwave Alert
```

Outcome:

The Heatwave Alert Stack System was successfully implemented using stack operations. Heatwave alerts based on temperature readings were stored as active alerts and removed using the Pop operation.

Conclusion:

The program demonstrates the use of Stack, Push, Pop, Display, conditional statements, and user input to manage active heatwave alerts based on temperature levels.